

(c) Complex numbers u and v are $u = q + 2i$ and $v = 1 - 2i$.
Given that $\left|\frac{u}{v}\right| = 13$, find all possible values of q .

$$\begin{aligned}\left|\frac{u}{v}\right| &= \left|\frac{q + 2i}{1 - 2i}\right| \\&= \left|\frac{(q + 2i)(1 + 2i)}{5}\right| \\&= \left|\frac{q + 2qi + 2i - 4}{5}\right| = 13 \\&\sqrt{\frac{(q - 4)^2}{25} + \frac{(2q + 2)^2}{25}} = 13 \\(q - 4)^2 + (2q + 2)^2 &= 4225 \\q^2 - 8q + 16 + 4q^2 + 8q + 4 &= 4225 \\5q^2 &= 4205 \\q^2 &= 841 \\q &= \pm 29\end{aligned}$$